

## Supplemental Online Content

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This supplemental material has been provided by the authors to give readers additional information about their work.

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## **Methods Supplementary Appendix**

### **Participants:**

Eligible participants for ARMMS-T2D included those who were originally randomized in one of the four original trials (STAMPEDE, SLIMM-T2D, TRIABETES or CROSSROADS) to bariatric surgery (Roux-en-Y gastric bypass [RYGB], sleeve gastrectomy [SG] or adjustable gastric banding [AGB]) or medical/lifestyle treatment and who underwent their allocated treatment intervention post-randomization. Other inclusion criteria from the parent studies were: (a) diagnosis of T2D, (b) BMI 27-45 kg/m<sup>2</sup>, (c) age 18-65 years, (d) candidate for general anesthesia, (e) ability to perform exercise and other components of the lifestyle intervention, and (f) negative urine pregnancy test at baseline (if applicable). T2D was confirmed by requiring diabetes medication and/or having a diabetes diagnosis according to American Diabetes Association criteria.<sup>1,2</sup> Sex/gender data were collected by self-report with options of male, female, other provided. Details of the original recruitment and randomization processes including the methods of allocation concealment for each of the four original trials are available in these publications.<sup>3-14</sup> None of the study teams were masked to group assignment.

All eligible participants were invited to continue long-term observation at their original site, and voluntary informed written consent was obtained for participation in the pooled consortium study. The study protocol was approved by the Institutional Review Boards at each site. A diagram depicting the flow of patients from the original trials into the ARMMS-T2D follow-up study is shown in **Figure 1**, main paper.

### **Procedures:**

Randomization occurred at each of the 4 sites independently and differed by site. At Cleveland Clinic, participants were randomized to surgery (RYGB or SG) vs. medical/lifestyle therapy; at the Joslin Diabetes Center/Brigham and Women's Hospital participants chose which surgical procedure they preferred and were then randomized to that procedure or to medical/lifestyle therapy; at the University of Pittsburgh participants were randomized to RYGB, AGB, or medical/lifestyle therapy; and at University of Washington/Kaiser Permanente Washington, to RYGB or medical/lifestyle. All surgical procedures were performed laparoscopically. Those randomized to medical/lifestyle received individualized nutrition counselling, instruction on exercise and self-monitoring of glucose, plus behavioral education and counselling for at least 1 year after randomization. The protocols for delivery of the medical/lifestyle intervention varied across sites and have been published with the original trials.<sup>3-14</sup> After the first year, all patients at University of Pittsburgh and Cleveland Clinic remained in a low level lifestyle intervention including exercise and nutritional counseling monthly and participants at Joslin Diabetes Center/Brigham and University of Washington/Kaiser Permanente Washington were encouraged to maintain participation in supervised or unsupervised exercise, as well as to continue behavioral and nutritional counseling.

### **Data collection and oversight:**

Annual study visits were scheduled within a 12-month window ( $\pm 6$  months) and centered at the anniversary date of randomization in their original trial. Participants were eligible for an annual follow up visit even if they had missed a prior visit. Blood and urine samples were collected and sent to the central laboratory (Northwest Lipid Research Laboratories at University of Washington until 05/2020, thereafter at Advanced Research and Diagnostic Laboratory at the University of Minnesota). Weight, waist circumference, systolic and diastolic blood pressure, and the use of glucose-lowering and other medications were obtained. Annual visits were scheduled at study centers. If in-person visits were not possible, remote visits could be performed as approved by the IRBs at individual sites. Telemedicine

visits and data extraction from electronic medical records were implemented to ensure continued engagement and patient safety during the COVID-19 pandemic. For any missed annual visits, laboratory and vital signs (except for fasting glucose) were also obtained from the medical charts. An Observational Study Monitoring Board (OSMB) appointed by NIH/NIDDK met twice a year to oversee the study. Because the follow up study was observational, there were no formal stopping rules based on safety.

### **Data harmonization:**

The primary outcome of the parent RCTs was based on change in HbA<sub>1c</sub>. Although the assays were performed at different laboratories, the efforts and success of the National Glycohemoglobin Standardization Program to standardize clinical laboratories against a universal calibrator and to reduce assay variability enabled the use of these baseline measures across centers. For measurement of height and weight, wall-mounted stadiometers and calibrated scales were used at each center. Blood pressure measurements were all obtained in the seated position after 5 minutes of rest, using the same arm for each participant.

### **Participant retention:**

At the initiation of the pooled study, 305 of the 316 patients who were originally randomized and underwent an intervention in one of the four original trials were available for follow-up (**Figure 1, main paper**). Participant retention strategy plans were crafted by the coordinators specific to the study populations in their specific geographical site. Specific strategies included telephone contact at 3-month intervals, evening and/or weekend options for study-related visits, newsletters to update participants on study progress, review of annual participant retention survey data, a study web site (<https://www.lerner.ccf.org/quantitative-health/armmst2d/about/>), monetary remuneration/reimbursement, and active outreach effort to maintain current contact information.

## **Statistical Supplementary Appendix**

### **Statistical analysis:**

Baseline participant characteristics were summarized as appropriate descriptive statistics for the medical/lifestyle group, bariatric surgery group and each of the three surgical procedures. The analyses followed the intention-to-treat (ITT) principle. The primary outcome of change in HbA<sub>1c</sub> was assessed using a linear mixed-effect model, which included group (medical/lifestyle vs. surgery), visit, their interaction, site, and baseline HbA<sub>1c</sub> as fixed effects. The models also included a random intercept at the participant level. The random intercept was assumed to follow a Gaussian distribution with a zero mean. Least-square means were derived at each visit by group, and the group comparison at each visit was performed by applying appropriate linear contrast to the model. The group comparison at year 7 was the primary comparison. A secondary analysis was conducted where the binary group assignment was replaced by a 4-category variable representing the three surgical procedures and the medical/lifestyle intervention, allowing pairwise comparisons among the four groups with multiplicity adjustment.

Secondary outcomes for longitudinal continuous variables such as the percent changes of weight and cardiometabolic risk factors (e.g., blood pressure and lipid parameters) were analyzed in the same manner as the primary outcome. Each model controlled for site and the baseline value as the covariates. For longitudinal binary outcomes such as diabetes remission, glycemic control and the use of medications for diabetes, the generalized estimating equation approach was used, where the models included group, visit, their interaction, site and baseline HbA<sub>1c</sub> as covariates. Least-square (LS) estimates were derived by visits and group, and the standard errors of the LS estimates were obtained using the sandwich method. The

results of the secondary outcomes were considered for hypothesis generating and were not adjusted for multiple testing.

The exploratory analyses for the subgroup of the participants with BMI <35 kg/m<sup>2</sup> at baseline were conducted using the same set of methods but were restricted to those with low baseline BMI.

To account for the crossovers from the medical/lifestyle group to the surgical group, the per-protocol (PP) analyses were also conducted using the inverse probability weighting (IPW) approach<sup>15-17</sup>. The study data were collected annually according to the study design. For the ITT analysis, annual visit was the only time-varying variable in the model, and group assignment was fixed as the treatment received at randomization for each patient. For the primary outcome of change in HbA<sub>1c</sub> at year  $t$ , the mean under the mixed-effects model described above can be expressed as  $E(Y_t) = \alpha_t + \beta_t X + \gamma^T Z$ , where  $X$  represents the group (=0 for medical/lifestyle and =1 for surgical intervention) and  $Z$  represents the covariates of site and baseline HbA<sub>1c</sub>. Note that the expectation of the random intercept in the mixed-effects model was zero.  $\alpha_t, \beta_t, t = 1, \dots, 12$ , and  $\gamma$  are the unknown parameters, where  $\beta_t = E(Y_t|X = 1) - E(Y_t|X = 0)$  represents the difference of the mean change in HbA<sub>1c</sub> (surgical group minus medical/lifestyle group) at year  $t$ . For the PP analysis, treatment group became a time-varying variable. More specifically, patients who crossed over were considered in the medical/lifestyle group before they crossed over and were in the surgical group after the crossover. Note that there were no crossovers from surgery to the medical/lifestyle intervention as the surgical procedures were irreversible. To model the time-varying treatment group, the model was adapted as  $E(Y_t) = \alpha_t + \beta_t X_t + \gamma^T Z$ . For a patient who crossed over to the surgical group between years  $s-1$  and  $s$  ( $s > 0$ ),  $X_t = 0$  for  $t < s$  and  $X_t = 1$  for  $t \geq s$ . HbA<sub>1c</sub> was designated as a confounder for the treatment assignment, and the IPW approach was applied in the PP analysis. The inverse probability weight at visit  $t$  was defined as  $w_t = \prod_{k=1}^t \frac{P(X_k|\bar{X}_{k-1})}{P(X_k|\bar{X}_{k-1}, \bar{Y}_{k-1}, Z)}$ <sup>18</sup>, where  $X_k$  represents the group assignment at visit  $k$ , and  $\bar{X}_{k-1}$  and  $\bar{Y}_{k-1}$  represent the treatment history and the HbA<sub>1c</sub> measurements up to visit  $k-1$ , respectively. For the term of  $\frac{P(X_k|\bar{X}_{k-1})}{P(X_k|\bar{X}_{k-1}, \bar{Y}_{k-1}, Z)}$  in  $w_t$ , the numerator was estimated from fitting a logistic regression model with  $X_k$  as the dependent variable and the previous group assignments as the predictors. The denominator was estimated from a logistic model that further included site and HbA<sub>1c</sub> as covariates. The  $w_t$ 's were stabilized weights, and the calculations were performed using the *ipw* package in R (cran.r-project.org) with truncations at the 2.5% and 97.5% percentiles. The PP analyses thus weighted each observation in the data with the inverse probability weight and fit a weighted linear mixed-effects model. Since the weights were estimated from the data, the robust sandwich method was used to derive the standard errors for the model parameters. This same method was applied to the PP analyses of weight change.

The use of the inverse probability of treatment weights essentially created a synthetic sample in which HbA<sub>1c</sub> was not confounding with the treatment assignment, which allows for the estimation of the average treatment effect as the target estimand<sup>19</sup>. The average treatment effect should be interpreted as the effect in the entire study population. The validity of the IPW approach relies on several key assumptions such as no unmeasured confounding, positivity, and correct specification of the IPW models<sup>18,20</sup>, and future research is desired for an in-depth investigation of these assumptions.

**Missing data:** Since the four original trials started at different times and there was also lagged patient enrollment within each trial, the lengths of the follow-up times varied by participants. All participants reached a minimum of 7-year follow-up and the range was from 7 to 15 years. Thus, some missing data were due to the shorter lengths of follow-up of certain participants. For example, the participants with only 7 years of follow-up would not have data at years 8 to 12. Missing clinical and laboratory data caused by missed study visits were recovered using the participants' electronic medical records from their routine clinical visits, if available. Death or lost to follow-up were assumed to be random censoring

events, and the monotone missing data caused by these events were also considered missing at random. Furthermore, the study outcomes were analyzed using the linear mixed-effects model or the generalized estimating equation method as described above, which accounted for the missingness property under the assumption of missing at random<sup>21,22</sup>.

Adverse events and study outcomes were summarized as percentages and were compared between the two groups using the chi-squared or Fisher's exact test. All analyses were performed using RStudio (Boston, MA) and SAS 9.4 (Cary, NC).

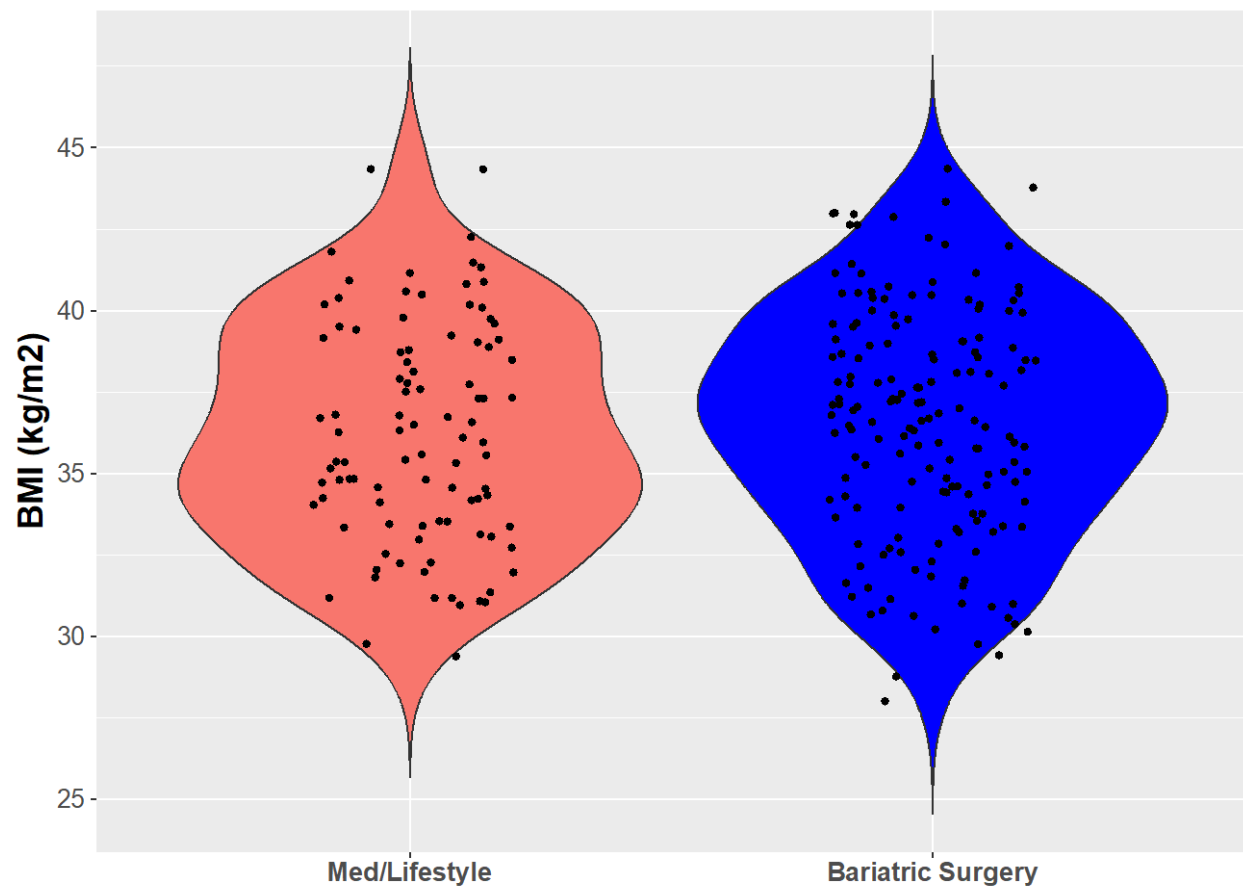
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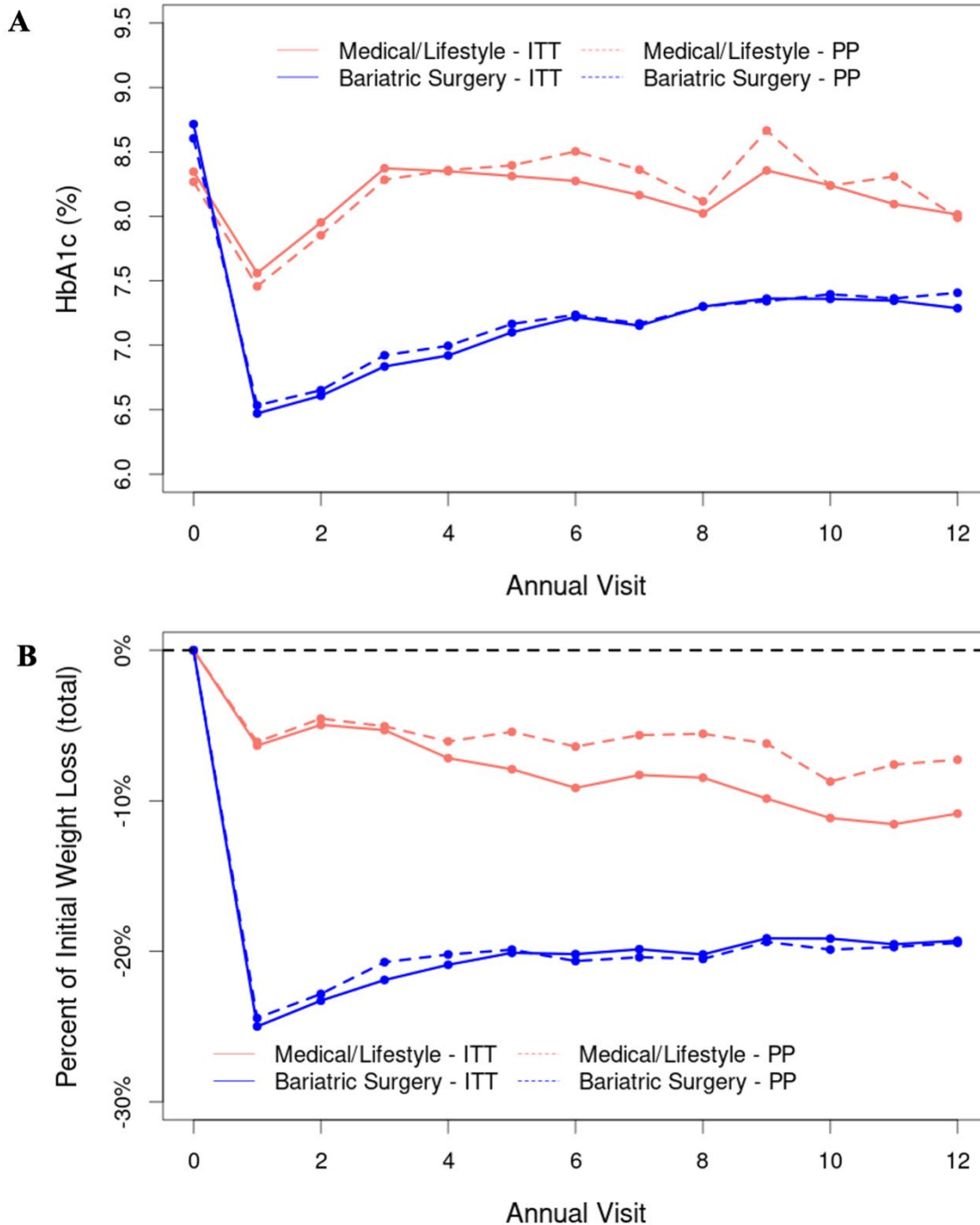
## Supplementary Figures

eFigure 1. Violin plot of baseline BMI by group





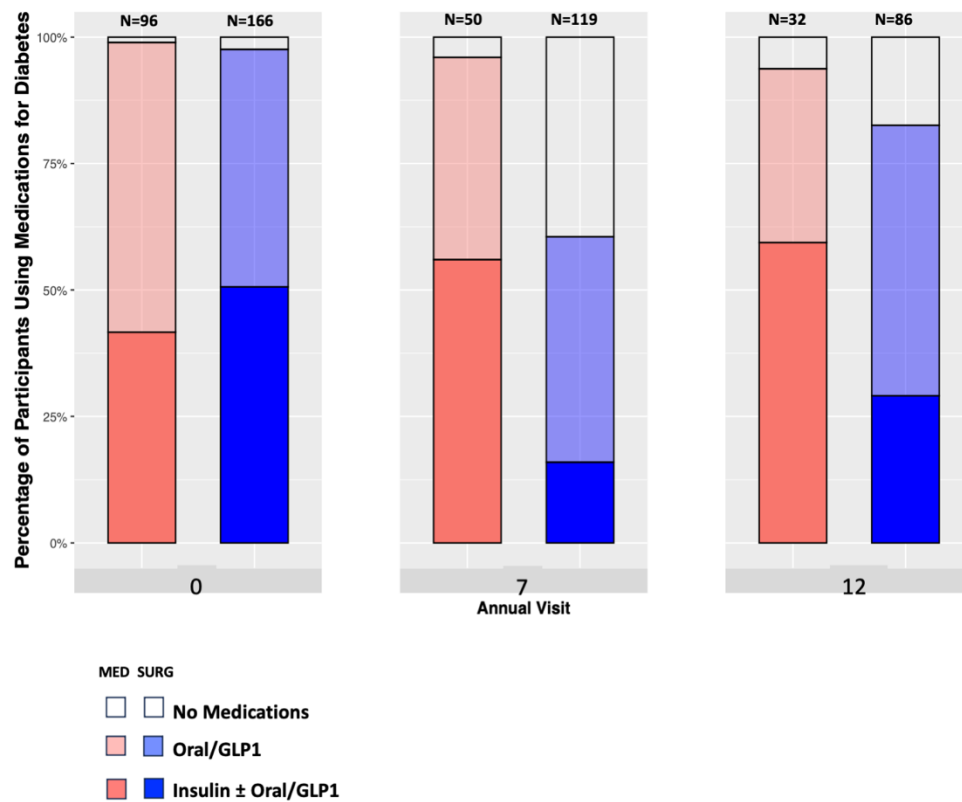
**eFigure 2. Comparison of per-protocol analysis and intention-to-treat analysis of HbA<sub>1c</sub> and weight change. A. HbA<sub>1c</sub> over time. B. Weight loss over time.** The solid lines show the results of intention-to-treat (ITT) analysis, and the dashed lines show the results of the per-protocol (PP) analysis. The per-protocol analysis accounted for the crossovers from medical/lifestyle group to surgery using the inverse probability weighting approach. Least-squared means are plotted over time.



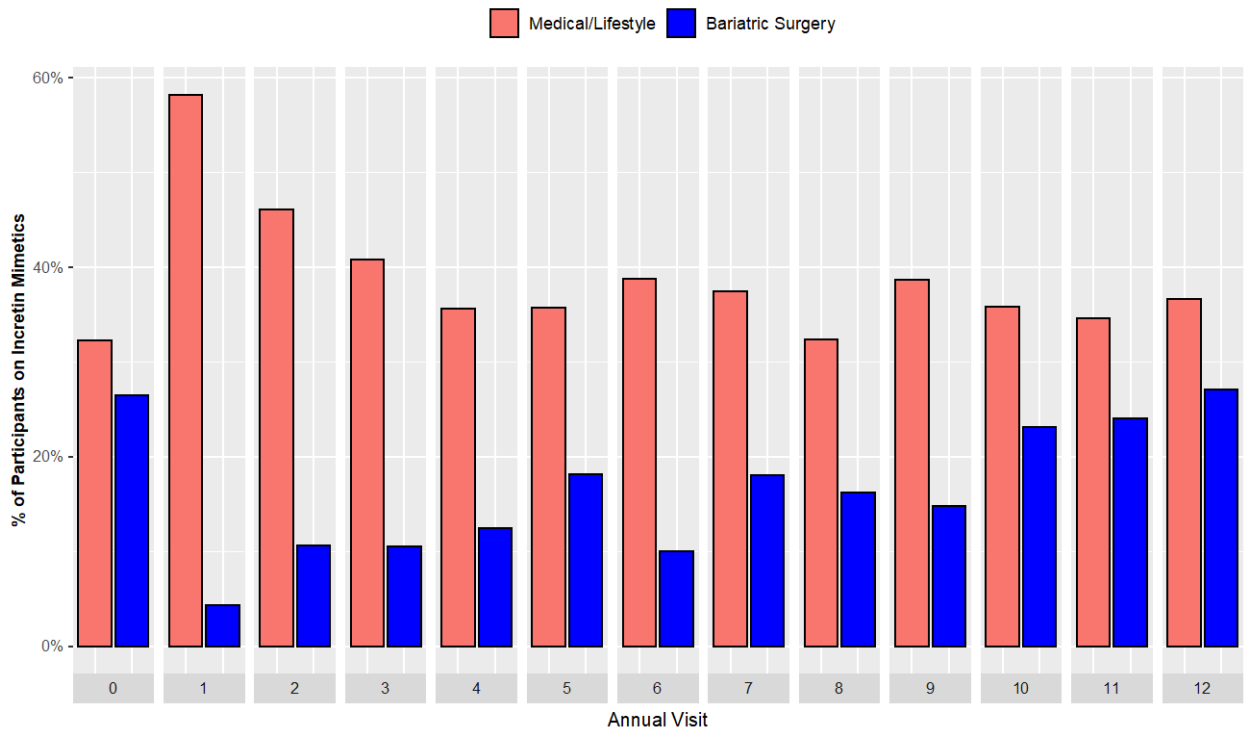
In the ITT analysis, weight loss at year 7 was 8.3% for the medical group and 19.9% for the surgical group, respectively. Using the PP analysis, weight loss at year 7 was 5.6% and 20.4% for the medical and surgical groups, respectively.

**eFigure 3. Use of medications for diabetes at baseline, 7 and 12 years**

Abbreviations: MED, medical/lifestyle group; SURG, Bariatric surgery.



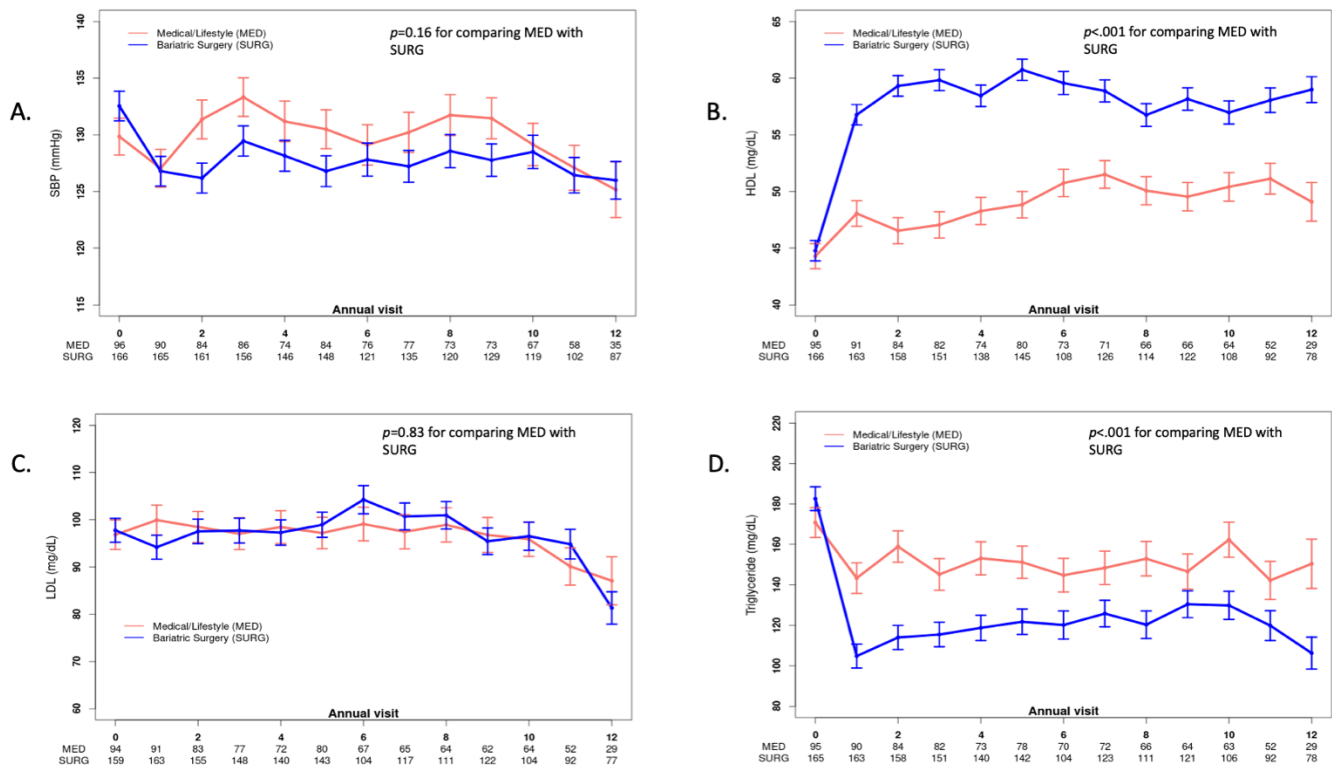
eFigure 4. Incretin medication use by group.



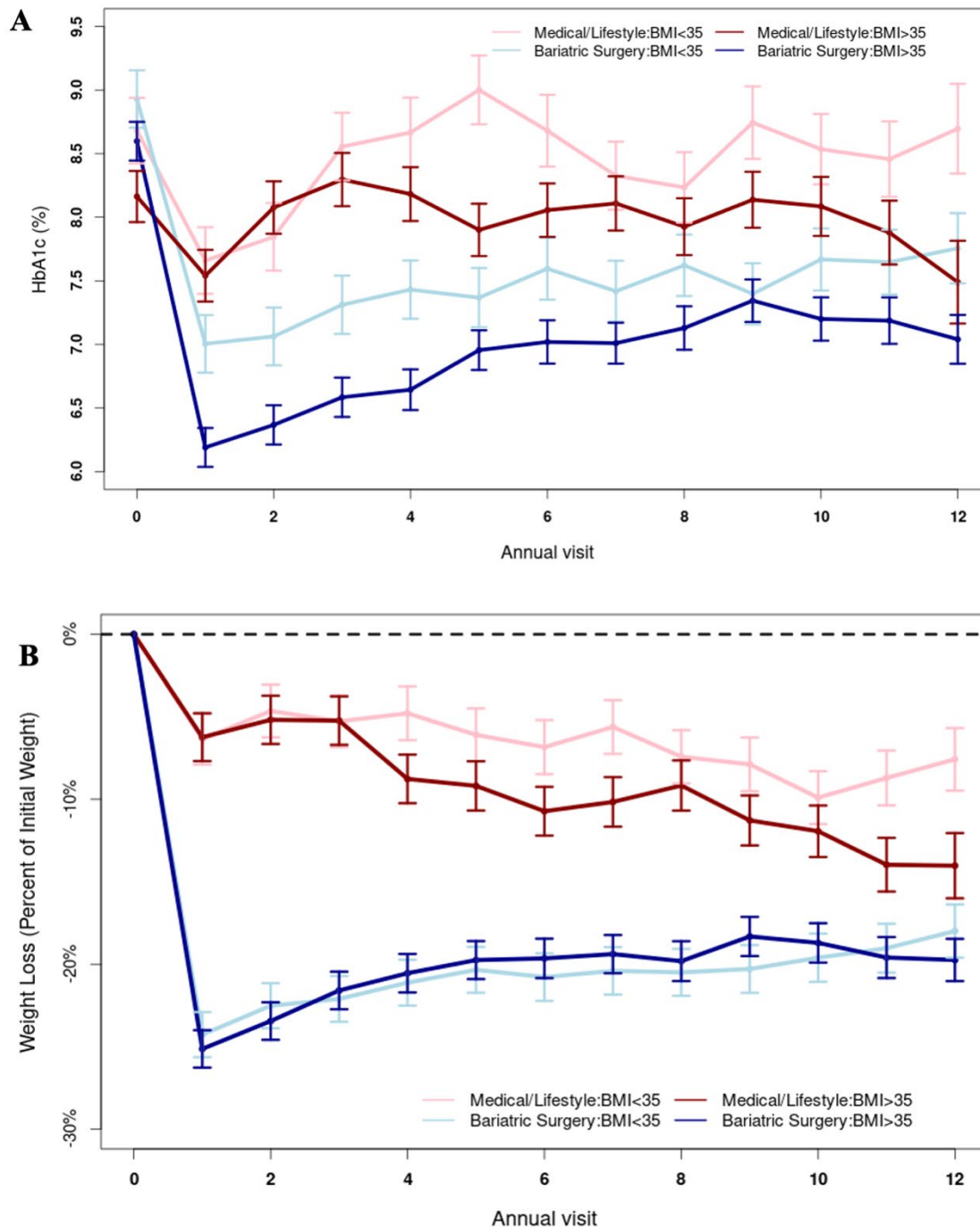
$p<0.001$  for the overall comparison between the two groups across all annual visits.

**eFigure 5. Change in systolic blood pressure and lipids over time by group. A. Systolic BP. B. HDL cholesterol. C. LDL cholesterol. D. Triglycerides.**

Least-square means and the standard errors are plotted.  $p=0.16$ ,  $<0.001$ ,  $0.83$ , and  $<0.001$ , respectively, for overall group comparisons. Abbreviations: MED, medical/lifestyle group; SURG, Bariatric surgery.



**eFigure 6. Subgroup analysis for participants with BMI <35 and >35 kg/m<sup>2</sup> at randomization. A. HbA<sub>1c</sub> over time. B. Weight loss over time.** Least-squared (LS) means and error bars (i.e., one standard error for the corresponding LS mean) are plotted over time by group.



At year 7, participants in the medical/lifestyle group who had BMI <35 kg/m<sup>2</sup> at baseline had a mean HbA<sub>1c</sub> of 8.3% and those in the surgical group with baseline BMI <35 kg/m<sup>2</sup> had a mean HbA<sub>1c</sub> of 7.4%. The mean HbA<sub>1c</sub> of the medical and surgical participants with baseline BMI >35 kg/m<sup>2</sup> was 8.1% and 7.0%, respectively. For weight loss, participants in the medical/lifestyle group with baseline BMI <35 kg/m<sup>2</sup> had a weight loss of 5.6%, while the surgical group with baseline BMI <35 kg/m<sup>2</sup> had a weight loss of 20.4%. Weight loss of medical and surgical participants with baseline BMI >35 kg/m<sup>2</sup> was 10.1% and 19.3%, respectively. The group difference was 14.8% for BMI <35 kg/m<sup>2</sup> and 9.2% for BMI >35 kg/m<sup>2</sup> (p=0.03).

## Supplementary Tables

**eTable 1. Baseline characteristics of participants who achieved 7 years of follow-up and 12 years of follow-up <sup>a</sup>**

|                                | With 7 years of follow-up<br>(N=262) <sup>b</sup> | With 12 years of follow-up<br>(N=130) <sup>b</sup> |
|--------------------------------|---------------------------------------------------|----------------------------------------------------|
| <b>Demographics:</b>           | n (%)                                             | n (%)                                              |
| Age (yrs)                      | 49.9 (8.3)                                        | 48.6 (7.5)                                         |
| Sex                            |                                                   |                                                    |
| Female                         | 179 (68.3)                                        | 93 (71.5)                                          |
| Male                           | 83 (31.7)                                         | 37 (28.5)                                          |
| Race                           |                                                   |                                                    |
| Black                          | 81 (30.9)                                         | 39 (30.0)                                          |
| White                          | 176 (67.2)                                        | 89 (68.5)                                          |
| Other                          | 4 (1.9)                                           | 2 (1.5)                                            |
| <b>Anthropometrics:</b>        |                                                   |                                                    |
| Waist (cm)                     | 114.6 (9.8)                                       | 113.8 (9.9)                                        |
| Weight (kg)                    | 104.3 (15.3)                                      | 102.1 (15.1)                                       |
| BMI (kg/m <sup>2</sup> )       | 36.4 (3.5)                                        | 36.1 (3.4)                                         |
| BMI<35 kg/m <sup>2</sup>       | 96 (36.6)                                         | 52 (40.0)                                          |
| Systolic BP (mmHg)             | 132.7 (17.1)                                      | 135.5 (17.4)                                       |
| Diastolic BP (mmHg)            | 80.1 (9.8)                                        | 80.7 (10.2)                                        |
| Diabetes Duration (yrs)        | 8.5 (5.4)                                         | 8.0 (5.3)                                          |
| <b>Laboratory:</b>             |                                                   |                                                    |
| HbA1c (%)                      | 8.5 (1.5)                                         | 8.8 (1.6)                                          |
| HbA1c<7%                       | 31 (11.8)                                         | 13 (10.0)                                          |
| Fasting Glucose (mg/dL)        | 166.4 (63.6)                                      | 176.0 (69.5)                                       |
| Total Cholesterol (mg/dL)      | 177.0 (42.7)                                      | 184.7 (45.5)                                       |
| HDL (mg/dL)                    | 43.4 (12.2)                                       | 44.5 (12.1)                                        |
| LDL (mg/dL)                    | 98.8 (33.8)                                       | 103.0 (35.1)                                       |
| Triglycerides (mg/dL)          | 144.0 [97.0, 229.0]                               | 154.0 [103.0, 236.0]                               |
| Serum Creatinine (mg/dL)       | 0.7 (0.2)                                         | 0.7 (0.2)                                          |
| Urine Albumin/Creatinine Ratio | 7.0 [4.0, 22.0]                                   | 8.0 [5.0, 22.8]                                    |
| <b>Medication use:</b>         |                                                   |                                                    |
| Statins                        | 192 (73.3)                                        | 94 (72.3)                                          |
| ACEi/ARB                       | 173 (66.0)                                        | 81 (62.3)                                          |
| ACEi                           | 137 (52.3)                                        | 60 (46.2)                                          |
| Insulin                        | 118 (45.0)                                        | 56 (43.1)                                          |
| Beta Blockers                  | 46 (17.6)                                         | 19 (14.6)                                          |
| ARB                            | 38 (14.5)                                         | 21 (16.2)                                          |

<sup>a</sup> Categorical variables are summarized as n (%). Continuous variables are summarized as mean (SD) and median [P25, P75]

<sup>b</sup> All participants in the study reached 7 years of follow-up and a subset of 130 participants reached 12 years of follow-up.

Abbreviations: BMI, body mass index; HDL, high density lipoprotein cholesterol; LDL, low density lipoprotein cholesterol, ACEi, angiotensin converting enzyme inhibitor; ARB, angiotensin receptor blocker

**eTable 2. Laboratory and clinical outcomes at year 12 and changes from baseline <sup>a</sup>**

|                                      | Medical/Lifestyle |                                   | Bariatric Surgery |                        | Group Comparison                  |         |
|--------------------------------------|-------------------|-----------------------------------|-------------------|------------------------|-----------------------------------|---------|
| Outcome                              | Year 12           | Change from baseline <sup>b</sup> | Year 12           | Change from baseline   | Difference in change <sup>c</sup> | P-value |
| HbA <sub>1c</sub> (%)                | 7.9 (1.4)         | -0.3 (-0.8,0.1)                   | 7.3 (1.5)         | -1.4 (-1.7,-1.1)       | -1.1 (-1.7,-0.5)                  | <.001   |
| Fasting glucose (mg/dL) <sup>d</sup> | 137.6 (45.4)      | -6% (-19.6%,7.6%)                 | 130.3 (47.2)      | -11.1% (-20%,-2.2%)    | -5.1% (-21%,10.9%)                | 0.53    |
| Weight (kg)                          | 92.8 (13.5)       | -10.8% (-13.5%,-8.2%)             | 83.7 (17.8)       | -19.3% (-21.3%,-17.3%) | -8.4% (-11.7%,-5.2%)              | <.001   |
| SBP (mmHg)                           | 125.6 (15.9)      | -4.7% (-8.5%,-1%)                 | 128.4 (15.8)      | -4.5% (-7.1%,-1.9%)    | 0.3% (-4.2%,4.7%)                 | 0.91    |
| DBP (mmHg)                           | 71.8 (10.2)       | -7.7% (-11.4%,-4%)                | 74.7 (10)         | -6.6% (-9.1%,-4.1%)    | 1.1% (-3.2%,5.5%)                 | 0.61    |
| LDL (mg/dL)                          | 90.9 (39.4)       | -7.9% (-19.9%,4%)                 | 87.6 (31.2)       | -11.9% (-20.2%,-3.7%)  | -4% (-18.2%,10.2%)                | 0.58    |
| HDL (mg/dL)                          | 50.5 (13.2)       | 16.6% (8.4%,24.7%)                | 57.7 (17.9)       | 38.1% (32.5%,43.7%)    | 21.5% (11.9%,31.2%)               | <.001   |
| Total cholesterol (mg/dL)            | 170.8 (48.1)      | -4.2% (-11.1%,2.6%)               | 168.9 (38.9)      | -5.5% (-10%,-0.9%)     | -1.2% (-9.3%,6.9%)                | 0.77    |
| Triglyceride (mg/dL)                 | 119 [89, 164]     | 9.8% (-4.9%,24.5%)                | 104 [75, 126.8]   | -23.3% (-33.3%,-13.2%) | -33.1% (-50.5%,-15.7%)            | <.001   |
| Serum creatinine (mg/dL)             | 0.9 (0.7)         | 29.5% (19.3%,39.7%)               | 0.9 (0.3)         | 20.3% (13.3%,27.3%)    | -9.2% (-21.2%,2.9%)               | 0.14    |
| Urine albumin to creatinine ratio    | 12.1 [6.4, 30.8]  | 2.2 (1.4, 3.6)                    | 8.1 [5.2, 20.1]   | 1.5 (1.1, 2.1)         | -0.7 (-1.9, 0.4)                  | 0.20    |
| Remission of Diabetes                | 0% <sup>e</sup>   | NA <sup>e</sup>                   | 12.7%             | 23.6 (1.2, 465.3)      | NA <sup>e</sup>                   | <.001   |
| Use of medications for diabetes      | 93.8%             | 0.06 (0.002, 2.16)                | 82.6%             | 0.10 (0.02, 0.39)      | 0.45 (0.15, 1.36)                 | 0.16    |
| Oral/GLP1 only                       | 34.4%             | 0.45 (0.23, 0.85)                 | 53.5%             | 1.14 (0.70, 1.85)      | 1.80 (0.85, 3.82)                 | 0.12    |
| Insulin and/or oral/GLP1             | 59.4%             | 2.31 (1.23, 4.32)                 | 29.1%             | 0.40 (0.25, 0.64)      | 0.25 (0.10, 0.58)                 | 0.001   |
| HbA <sub>1c</sub> <7.0%              | 29.4%             | 3.15 (1.20, 8.30)                 | 54.6%             | 6.57 (3.56, 12.1)      | 2.90 (1.15, 7.29)                 | 0.02    |
| HbA <sub>1c</sub> <6.5%              | 19.6%             | 2.68 (0.92, 7.81)                 | 40.0%             | 4.90 (2.54, 9.48)      | 2.75 (1.09, 6.90)                 | 0.03    |

<sup>a</sup> Mean (SD) or median [inter-quartile range] presented for baseline and year-7 data, and least-square estimate (95% CI) presented for the changes and group comparisons

<sup>b</sup> Net change presented for HbA<sub>1c</sub>, fold change (i.e., ratio of year 7 data over baseline data) for urine albumin and creatinine ratio, relative changes (i.e., percentage change from baseline) for other numeric outcomes, and odds ratios (7-year over baseline) for binary outcomes (remission, glycemic control and use of medications for diabetes)

<sup>c</sup> For numeric outcomes, the difference is defined as surgery group's 7-year change minus the 7-year change of the medical/lifestyle group; for binary outcomes, the difference is the odds ratio at year 7 (the odds of 7-year outcome in the surgery group over the medical/lifestyle group).

<sup>d</sup> Only included fasting glucose measurements obtained at in-person study visits.

<sup>e</sup> The 12-year remission rate was  $2e^{-16}$  in the medical group, and thus the odds ratio was close to 0 within the medical group and almost infinity for the group comparison.

Abbreviations: SBP, systolic blood pressure; DBP, diastolic blood pressure, HDL, high density lipoprotein cholesterol; LDL, low density lipoprotein cholesterol



**eTable 3. Frequency of nutritional abnormalities in the bariatric surgery group, individual surgery subgroups, and medical group through 12 years**

| Nutritional Abnormality       | Definition of Abnormality             | Study Group |    |                |       | p-value <sup>a</sup> |
|-------------------------------|---------------------------------------|-------------|----|----------------|-------|----------------------|
|                               |                                       |             | n  | N <sup>b</sup> | %     |                      |
| <b>Anemia</b>                 | Hemoglobin <11.5 g/dL                 | <b>MED</b>  | 11 | 87             | 12.6% | --                   |
|                               |                                       | <b>SURG</b> | 35 | 147            | 23.8% | 0.04                 |
|                               |                                       | <i>RYGB</i> | 25 | 78             | 32.1% | 0.003                |
|                               |                                       | <i>SG</i>   | 8  | 36             | 22.2% | 0.18                 |
|                               |                                       | <i>AGB</i>  | 2  | 33             | 6.1%  | 0.30                 |
| <b>Low Iron</b>               | Iron <59 µg/dL or <6.6 µmol/L         | <b>MED</b>  | 15 | 88             | 17.0% | --                   |
|                               |                                       | <b>SURG</b> | 48 | 151            | 31.8% | 0.01                 |
|                               |                                       | <i>RYGB</i> | 30 | 81             | 37.0% | 0.003                |
|                               |                                       | <i>SG</i>   | 10 | 37             | 27.0% | 0.20                 |
|                               |                                       | <i>AGB</i>  | 8  | 33             | 24.2% | 0.37                 |
| <b>Vitamin B12 deficiency</b> | Vitamin B12 <230 pg/mL                | <b>MED</b>  | 8  | 88             | 9.1%  | --                   |
|                               |                                       | <b>SURG</b> | 20 | 151            | 13.2% | 0.34                 |
|                               |                                       | <i>RYGB</i> | 15 | 81             | 18.5% | 0.07                 |
|                               |                                       | <i>SG</i>   | 4  | 37             | 10.8% | 0.77                 |
|                               |                                       | <i>AGB</i>  | 1  | 33             | 3.0%  | 0.26                 |
| <b>Vitamin D deficiency</b>   | Vitamin D2 (1,25 Dihydroxy) ≤20 ng/mL | <b>MED</b>  | 23 | 88             | 26.1% | --                   |
|                               |                                       | <b>SURG</b> | 35 | 151            | 23.2% | 0.61                 |
|                               |                                       | <i>RYGB</i> | 22 | 81             | 27.2% | 0.88                 |
|                               |                                       | <i>SG</i>   | 2  | 37             | 5.4%  | 0.008                |
|                               |                                       | <i>AGB</i>  | 11 | 33             | 33.3% | 0.43                 |
| <b>Hypocalcemia</b>           | Calcium <8.4 mg/dL                    | <b>MED</b>  | 0  | 88             | 0     | --                   |
|                               |                                       | <b>SURG</b> | 1  | 151            | 0.7%  | 0.44                 |
|                               |                                       | <i>RYGB</i> | 1  | 81             | 1.2%  | 0.30                 |
|                               |                                       | <i>SG</i>   | 0  | 37             | 0     | NA                   |
|                               |                                       | <i>AGB</i>  | 0  | 33             | 0     | NA                   |
| <b>Elevated PTH</b>           | PTH (intact) > 65 pg/mL               | <b>MED</b>  | 29 | 88             | 33.0% | --                   |
|                               |                                       | <b>SURG</b> | 66 | 151            | 43.7% | 0.10                 |
|                               |                                       | <i>RYGB</i> | 41 | 81             | 50.6% | 0.02                 |
|                               |                                       | <i>SG</i>   | 12 | 37             | 32.4% | 0.95                 |
|                               |                                       | <i>AGB</i>  | 13 | 33             | 39.4% | 0.51                 |
| <b>Hypoalbuminemia</b>        | Albumin <3.5 g/dL                     | <b>MED</b>  | 9  | 88             | 10.2% | --                   |
|                               |                                       | <b>SURG</b> | 23 | 151            | 15.2% | 0.27                 |
|                               |                                       | <i>RYGB</i> | 14 | 81             | 17.3% | 0.18                 |
|                               |                                       | <i>SG</i>   | 5  | 37             | 13.5% | 0.60                 |
|                               |                                       | <i>AGB</i>  | 4  | 33             | 12.1% | 0.76                 |
| <b>Severe hypoalbuminemia</b> | Albumin <2.8 g/dL                     | <b>MED</b>  | 1  | 88             | 1.1%  | --                   |
|                               |                                       | <b>SURG</b> | 0  | 151            | 0     | 0.19                 |

<sup>a</sup> Compared to the medical/lifestyle group; <sup>b</sup> Number of participants with observed data. Abbreviations: MED, medical/lifestyle group; SURG, Bariatric surgery; RYGB, Roux-en-Y gastric bypass; SG, sleeve gastrectomy; AGB, adjustable gastric band

**eTable 4. Least-square means of nutritional variables by group through 12 years**

|                     | Medical/Lifestyle |       | Bariatric Surgery |      | Difference <sup>a</sup> |       |         |
|---------------------|-------------------|-------|-------------------|------|-------------------------|-------|---------|
|                     | Mean              | SE    | Mean              | SE   | Estimate                | SE    | P-value |
| Hemoglobin (g/dL)   | 13.7              | 0.2   | 13                | 0.1  | -0.8                    | 0.2   | <.001   |
| Iron (µg/dL)        | 83.9              | 3.2   | 85.3              | 2.5  | 1.4                     | 4.1   | 0.74    |
| Vitamin B12 (pg/mL) | 760.8             | 105.1 | 1173.2            | 80.6 | 412.4                   | 132.4 | 0.002   |
| Vitamin D (ng/mL)   | 35.2              | 1.8   | 40.5              | 1.4  | 5.3                     | 2.3   | 0.02    |
| Calcium (mg/dL)     | 9.6               | 0.04  | 9.5               | 0.03 | -0.1                    | 0.0   | 0.03    |
| PTH (pg/mL)         | 51.1              | 3.3   | 60.3              | 2.5  | 9.2                     | 4.2   | 0.03    |
| Albumin (g/dL)      | 4.1               | 0.03  | 4.1               | 0.02 | -0.01                   | 0.04  | 0.72    |

<sup>a</sup> Difference: the surgical mean minus the medical mean.

**eTable 5. Adverse events through 12 years<sup>a</sup>**

| Adverse Events                               | Medical/Lifestyle<br>(N=96) | Bariatric Surgery<br>(N=166) | p-value <sup>b</sup> |
|----------------------------------------------|-----------------------------|------------------------------|----------------------|
| <b>Cardiovascular and pulmonary</b>          |                             |                              |                      |
| Chest pain                                   | 10 (10.4)                   | 15 (9.0)                     | 0.71                 |
| Hypertension (new or severe exacerbation)    | 2 (2.1)                     | 6 (3.6)                      | 0.71                 |
| Diabetes related foot ulcers                 | 1 (1.0)                     | 4 (2.4)                      | 0.65                 |
| Tachycardia                                  | 3 (3.1)                     | 3 (1.8)                      | 0.49                 |
| Hypotension                                  | 4 (4.2)                     | 2 (1.2)                      | 0.20                 |
| Sleep apnea (new or severe exacerbation)     | 1 (1.0)                     | 2 (1.2)                      | 1.00                 |
| Carotid vascular disease                     | 2 (2.1)                     | 1 (0.6)                      | 0.56                 |
| <b>Gastrointestinal</b>                      |                             |                              |                      |
| Nausea/vomiting                              | 15 (15.6)                   | 43 (25.9)                    | 0.054                |
| Abdominal pain <sup>c</sup>                  | 10 (10.4)                   | 37 (22.3)                    | 0.02                 |
| Gastroesophageal reflux disease <sup>d</sup> | 10 (10.4)                   | 29 (17.5)                    | 0.12                 |
| Diarrhea                                     | 7 (7.3)                     | 20 (12.0)                    | 0.22                 |
| Constipation                                 | 10 (10.4)                   | 17 (10.2)                    | 0.96                 |
| Dysphagia <sup>e</sup>                       | 0 (0.0)                     | 12 (7.2)                     | 0.005                |
| Hiatal hernia                                | 2 (2.1)                     | 7 (4.2)                      | 0.49                 |
| Dumping syndrome <sup>f</sup>                | 0 (0.0)                     | 8 (4.8)                      | 0.03                 |
| Colon polyps                                 | 6 (6.2)                     | 7 (4.2)                      | 0.46                 |
| Gastrointestinal stricture                   | 0 (0.0)                     | 4 (2.4)                      | 0.30                 |
| Hematochezia                                 | 1 (1.0)                     | 3 (1.8)                      | 1.00                 |
| Colitis                                      | 0 (0.0)                     | 3 (1.8)                      | 0.30                 |
| Gastric prolapse <sup>g</sup>                | 0 (0.0)                     | 2 (1.2)                      | 0.53                 |
| Gastroparesis                                | 4 (4.2)                     | 0 (0.0)                      | 0.02                 |
| <b>Neurologic and Mental Health</b>          |                             |                              |                      |
| Vertigo/dizziness                            | 9 (9.4)                     | 27 (16.3)                    | 0.12                 |
| Depression                                   | 14 (14.6)                   | 24 (14.5)                    | 0.98                 |
| Concussion/contusion/head Injury             | 2 (2.1)                     | 13 (7.8)                     | 0.058                |
| Neuropathy                                   | 7 (7.3)                     | 11 (6.6)                     | 0.84                 |
| Memory loss/cognitive dysfunction            | 3 (3.1)                     | 6 (3.6)                      | 0.83                 |
| <b>Ocular</b>                                |                             |                              |                      |
| Cataracts                                    | 12 (12.5)                   | 20 (12.0)                    | 0.91                 |
| Blurred or double vision                     | 7 (7.3)                     | 11 (6.6)                     | 0.84                 |
| Glaucoma                                     | 5 (5.2)                     | 4 (2.4)                      | 0.23                 |
| <b>Infectious</b>                            |                             |                              |                      |
| Upper respiratory infection                  | 18 (18.8)                   | 43 (25.9)                    | 0.19                 |
| Pneumonia                                    | 3 (3.1)                     | 11 (6.6)                     | 0.23                 |
| Cellulitis                                   | 5 (5.2)                     | 6 (3.6)                      | 0.53                 |

| Adverse Events                                  | Medical/Lifestyle<br>(N=96) | Bariatric Surgery<br>(N=166) | p-value <sup>b</sup> |
|-------------------------------------------------|-----------------------------|------------------------------|----------------------|
| Systemic infection/sepsis                       | 0 (0.0)                     | 2 (1.2)                      | 0.53                 |
| Osteomyelitis                                   | 1 (1.0)                     | 1 (0.6)                      | 1.00                 |
| COVID-19                                        | 1 (1.0)                     | 1 (0.6)                      | 1.00                 |
| <b>Metabolic</b>                                |                             |                              |                      |
| Severe hyperglycemia                            | 8 (8.3)                     | 14 (8.4)                     | 0.98                 |
| Hyperparathyroidism                             | 0 (0.0)                     | 2 (1.2)                      | 0.53                 |
| <b>Fluid and electrolytes</b>                   |                             |                              |                      |
| Dehydration                                     | 2 (2.1)                     | 14 (8.4)                     | 0.058                |
| Hypokalemia                                     | 2 (2.1)                     | 8 (4.8)                      | 0.33                 |
| Hyperkalemia                                    | 1 (1.0)                     | 0 (0.0)                      | 0.37                 |
| Hypercalcemia                                   | 0 (0.0)                     | 1 (0.6)                      | 1.00                 |
| <b>Skeletal</b>                                 |                             |                              |                      |
| Back pain                                       | 18 (18.8)                   | 36 (21.7)                    | 0.57                 |
| Osteoarthritis                                  | 8 (8.3)                     | 6 (3.6)                      | 0.10                 |
| <b>Renal</b>                                    |                             |                              |                      |
| Renal insufficiency                             | 2 (2.1)                     | 4 (2.4)                      | 1.00                 |
| Acute renal failure                             | 1 (1.0)                     | 2 (1.2)                      | 1.00                 |
| <b>Miscellaneous</b>                            |                             |                              |                      |
| Allergy/intolerance to medication or supplement | 5 (5.2)                     | 9 (5.4)                      | 0.94                 |
| Alcohol abuse                                   | 2 (2.1)                     | 7 (4.2)                      | 0.49                 |
| Abdominal wall hernia                           | 1 (1.0)                     | 6 (3.6)                      | 0.43                 |

<sup>a</sup> Data are presented as n (%)

<sup>b</sup> Chi-squared test or Fisher's exact test.

<sup>c</sup> Abdominal pain occurred in 34.1% after sleeve gastrectomy (SG), 22.5% after Roux-en-Y gastric bypass (RYGB), and 8.3% after adjustable gastric banding (AGB).

<sup>d</sup> GERD occurred in 36.6% after SG, 11.2% after RYGB, and 11.1% after AGB.

<sup>e</sup> Dysphagia occurred in 12.2% after SG, 8.3% after AGB, and 4.5% after RYGB.

<sup>f</sup> Dumping syndrome occurred in 7.9% after RYGB and 2.4% after SG.

<sup>g</sup> Gastric prolapse after AGB.

**eTable 6. Surgical and endoscopic procedures through 12 years<sup>a</sup>**

| <b>Procedure</b>                         | <b>Medical/Lifestyle<br/>(N=96)</b> | <b>Bariatric Surgery<br/>(N=166)</b> | <b>p-value<sup>b</sup></b> |
|------------------------------------------|-------------------------------------|--------------------------------------|----------------------------|
| <b>Abdominal Surgery</b>                 |                                     |                                      |                            |
| Crossovers, conversions, and revisions   | 24 (25.0)                           | 15 (9.0)                             | <.001                      |
| Cholecystectomy                          | 2 (2.1)                             | 7 (4.2)                              | 0.49                       |
| Lysis of adhesions                       | 0 (0.0)                             | 2 (1.2)                              | 0.53                       |
| Hernia repair                            | 0 (0.0)                             | 2 (1.2)                              | 0.53                       |
| Exploratory laparoscopy                  | 0 (0.0)                             | 2 (1.2)                              | 0.53                       |
| Exploratory laparotomy                   | 0 (0.0)                             | 1 (0.6)                              | 1.00                       |
| Appendectomy                             | 0 (0.0)                             | 2 (1.2)                              | 0.53                       |
| Colon resection                          | 0 (0.0)                             | 2 (1.2)                              | 0.53                       |
| Hysterectomy                             | 0 (0.0)                             | 2 (1.2)                              | 0.53                       |
| Other abdominal surgery                  | 2 (2.1)                             | 5 (3.0)                              | 1.00                       |
| <b>Endoscopy</b>                         |                                     |                                      |                            |
| Endoscopic treatment of leak             | 0 (0.0)                             | 1 (0.6)                              | 1.00                       |
| Endoscopic treatment of dilated GJ stoma | 0 (0.0)                             | 2 (1.2)                              | 0.53                       |
| Endoscopic treatment of GI bleeding      | 1 (1.0)                             | 0 (0.0)                              | 0.37                       |
| Other therapeutic endoscopy              | 2 (2.1)                             | 5 (3.0)                              | 1.00                       |
| Other endoscopy                          | 3 (3.1)                             | 14 (8.4)                             | 0.09                       |
| <b>Orthopedic Procedures</b>             |                                     |                                      |                            |
| Joint injection                          | 9 (9.4)                             | 6 (3.6)                              | 0.053                      |
| Procedures for fractures                 | 1 (1.0)                             | 3 (1.8)                              | 1.00                       |
| Joint replacement                        | 5 (5.2)                             | 3 (1.8)                              | 0.12                       |
| Amputation                               | 0 (0.0)                             | 1 (0.6)                              | 1.00                       |
| Other orthopedic surgery                 | 13 (13.5)                           | 33 (19.9)                            | 0.19                       |
| <b>Eye Surgery</b>                       |                                     |                                      |                            |
| Cataract surgery                         | 8 (8.3)                             | 13 (7.8)                             | 0.89                       |
| Other eye surgery                        | 4 (4.2)                             | 6 (3.6)                              | 0.82                       |
| Vitrectomy                               | 2 (2.1)                             | 1 (0.6)                              | 0.56                       |
| <b>Plastic Surgery</b>                   |                                     |                                      |                            |
| Panniculectomy or abdominoplasty         | 0 (0.0)                             | 5 (3.0)                              | 0.16                       |
| Other plastic surgery                    | 2 (2.1)                             | 1 (0.6)                              | 0.56                       |

<sup>a</sup> n (%) presented; <sup>b</sup> Chi-squared test or Fisher's exact test

**eTable 7. Crossovers, conversions, and revisions through 12 years**

| <b>Operations</b>                                             | <b>n/N (%)</b> |
|---------------------------------------------------------------|----------------|
| <b>Crossovers:</b> from medical group to surgery <sup>a</sup> | 24/96 (25%)    |
| Medical group to AGB                                          | 1/96 (1%)      |
| Medical group to SG                                           | 15/96 (16%)    |
| Medical group to RYGB                                         | 8/96 (8%)      |
| <b>Conversions:</b> After AGB                                 |                |
| Band removal                                                  | 7/36 (19%)     |
| AGB to RYGB                                                   | 4/36 (11%)     |
| AGB to SG                                                     | 1/36 (3%)      |
| <b>Revisions:</b> SG to RYGB                                  | 3/41 (7%)      |

<sup>a</sup> Median time from randomization to crossover: 4.5 years, range: [0.4, 9.8].

Abbreviations: RYGB, Roux-en-Y gastric bypass; SG, sleeve gastrectomy; AGB, adjustable gastric band